Supplementary Material

Supplementary table 1: definition of dysplasia used by pathologist reading WATS samples.

Study	Dysplasia reported	Both positive	Only FB +	Only WATS +	Both negative
Anandasabapathy 2011	LGD/HGD/AC	15	23	16	97
Johanson 2011	LGD/HGD/AC	4	8	7	372
Smith 2019	HGD/AC	1	2	6	2478
Bisschops 2020	HGD/AC	25	10	14	98
Dunkle 2020	HGD/AC	2	5	1	94
Vennalaganti 2018	HGD/AC	6	7	29	70

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Supplementary material

Supplementary table 2: tabular presentation for QUADAS-2 results

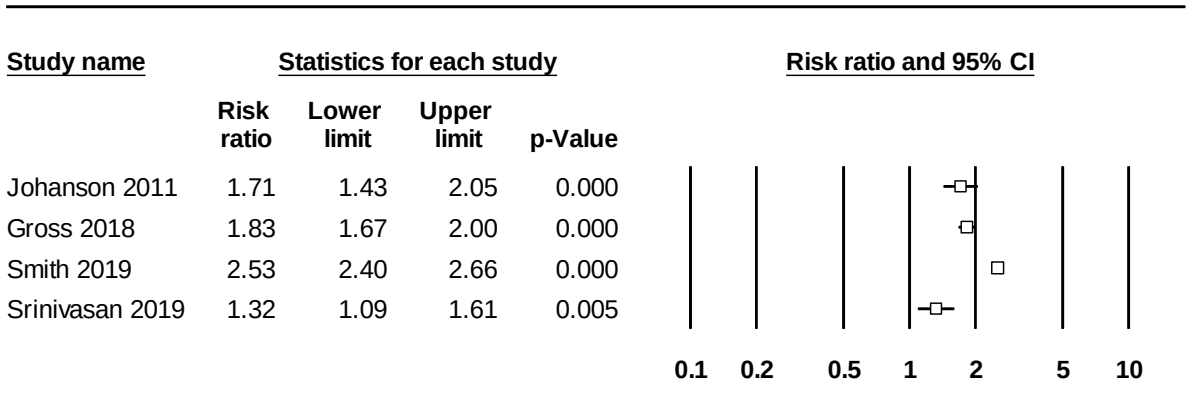
Study	Publication type	RISK OF BIAS				APPLICABILITY CONCERNS		
		PATIENT SELECTION	INDEX TEST	REFERENCE STANDARD	FLOW AND TIMING	PATIENT SELECTION	INDEX TEST	REFERENCE STANDARD
Vennalaganti 2018	Manuscript	☺	☺	☺	☺	☺	☺	☺
Anandasabapathy 2011	Manuscript	?	☺	☺	☺	☹	☺	☺
Johanson 2011	Manuscript	☺	☺	☺	☺	☺	☺	?
Gross 2018	Manuscript	☺	?	?	☺	☹	☺	?
Smith 2019	Manuscript	☺	☺	☺	☺	?	☺	☺
Raphael 2019	Manuscript	☺	☺	☺	?	☺	☺	☹

☺ Low Risk ☹ High Risk ? Unclear Risk

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Supplementary material

Supplementary figure 1: Forest plots of the additional yield in Barrett's esophagus (BE) detection

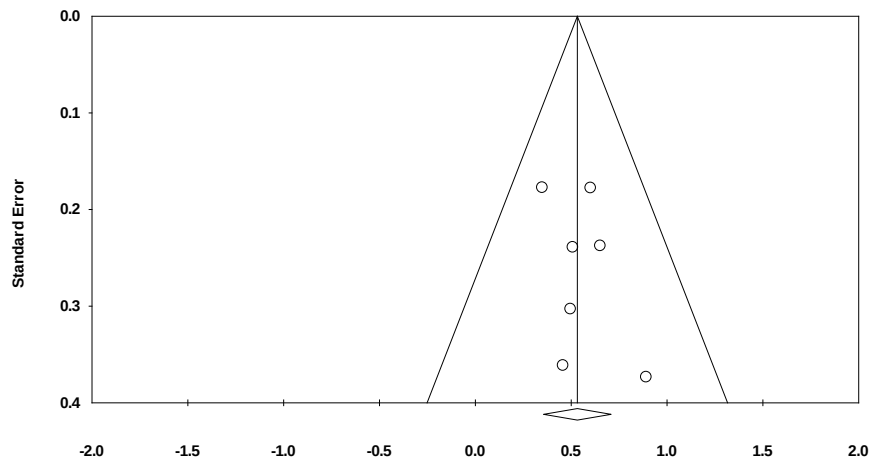


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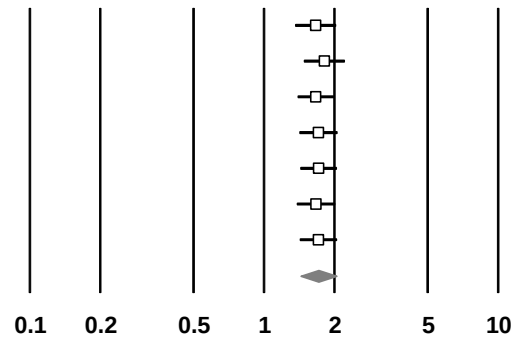
Supplementary figure 2: (a) funnel plot of standard of error by Log risk ratio for studies assessing the additional yield of all dysplasia in WATS compared to forceps biopsies; (b) forest plot of the additional yield of all dysplasia with one study removed at a time; (c) sensitivity analysis of the additional yield of all dysplasia when an outlier study was included; (d) sensitivity analysis of the additional yield of all dysplasia stratified by study design; and (e) sensitivity analysis of the additional yield of all dysplasia stratified by publication type (abstract vs. manuscript).

(a)



(b)

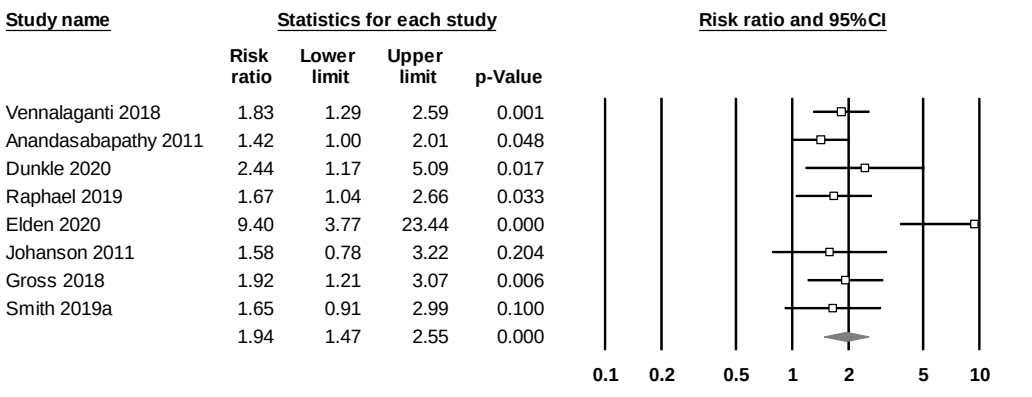
Study name	Statistics with study removed			
	Point	Lower limit	Upper limit	p-Value
Vennalaganti 2018	1.66	1.35	2.04	0.000
Anandasabapathy 2011	1.81	1.48	2.22	0.000
Dunkle 2020	1.67	1.39	2.00	0.000
Raphael 2019	1.71	1.41	2.07	0.000
Johanson 2011	1.71	1.43	2.05	0.000
Gross 2018	1.67	1.38	2.02	0.000
Smith 2019a	1.71	1.42	2.05	0.000
	1.70	1.43	2.03	0.000



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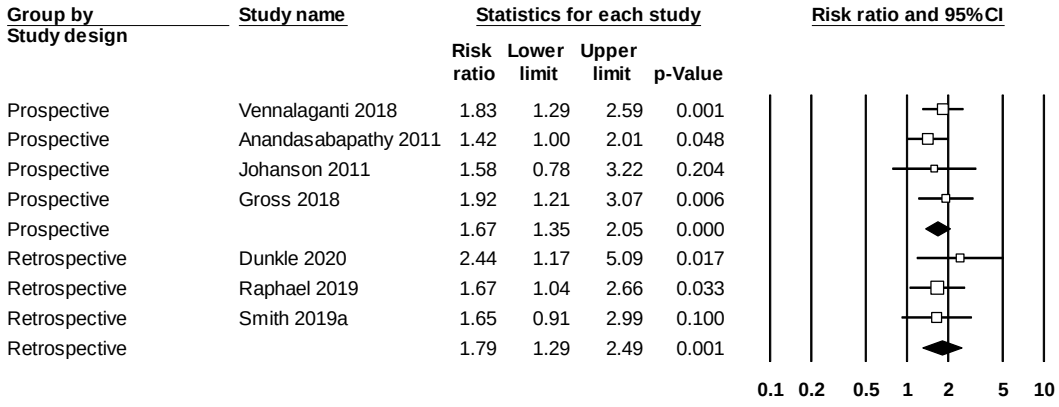
Supplementary material

(c)



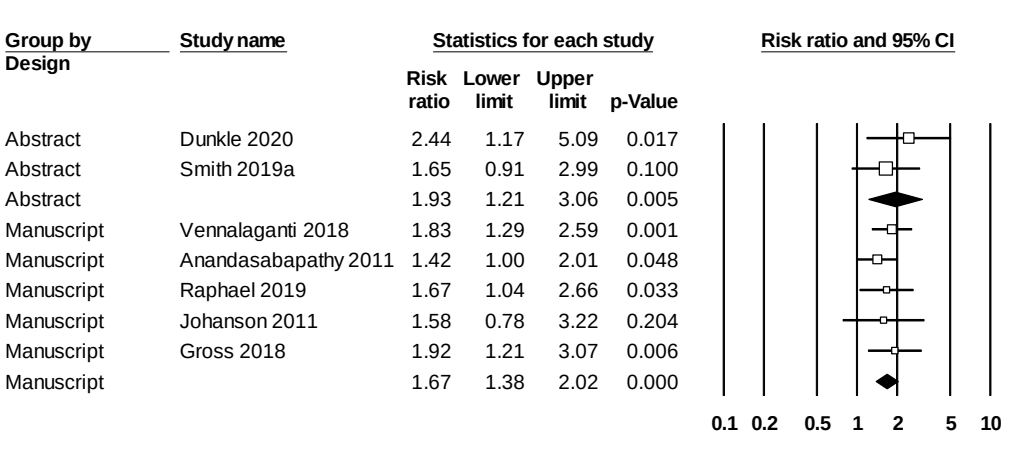
Study name	Risk ratio	Lower limit	Upper limit	p-Value
Vennalaganti 2018	1.83	1.29	2.59	0.001
Anandasabapathy 2011	1.42	1.00	2.01	0.048
Dunkle 2020	2.44	1.17	5.09	0.017
Raphael 2019	1.67	1.04	2.66	0.033
Elden 2020	9.40	3.77	23.44	0.000
Johanson 2011	1.58	0.78	3.22	0.204
Gross 2018	1.92	1.21	3.07	0.006
Smith 2019a	1.65	0.91	2.99	0.100
	1.94	1.47	2.55	0.000

(d)



Group by Study design	Study name	Risk ratio	Lower limit	Upper limit	p-Value
Prospective	Vennalaganti 2018	1.83	1.29	2.59	0.001
Prospective	Anandasabapathy 2011	1.42	1.00	2.01	0.048
Prospective	Johanson 2011	1.58	0.78	3.22	0.204
Prospective	Gross 2018	1.92	1.21	3.07	0.006
Prospective		1.67	1.35	2.05	0.000
Retrospective	Dunkle 2020	2.44	1.17	5.09	0.017
Retrospective	Raphael 2019	1.67	1.04	2.66	0.033
Retrospective	Smith 2019a	1.65	0.91	2.99	0.100
Retrospective		1.79	1.29	2.49	0.001

(e)



Group by Design	Study name	Risk ratio	Lower limit	Upper limit	p-Value
Abstract	Dunkle 2020	2.44	1.17	5.09	0.017
Abstract	Smith 2019a	1.65	0.91	2.99	0.100
Abstract		1.93	1.21	3.06	0.005
Manuscript	Vennalaganti 2018	1.83	1.29	2.59	0.001
Manuscript	Anandasabapathy 2011	1.42	1.00	2.01	0.048
Manuscript	Raphael 2019	1.67	1.04	2.66	0.033
Manuscript	Johanson 2011	1.58	0.78	3.22	0.204
Manuscript	Gross 2018	1.92	1.21	3.07	0.006
Manuscript		1.67	1.38	2.02	0.000

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Supplementary material

Appendix 1: search strategy and terms for each search engine.

Cochrane Library (Wiley Online Library)/CENTRAL

***

Search Name: Qumseya Barrett's Esophagus, WATS

ID	Search	Hits
#1	MeSH descriptor: [Barrett Esophagus] explode all trees	
#2	(barrett esophag* OR barrett's esophag* OR barretts esophag* OR barrett oesophag* OR barrett's oesophag* OR barretts oesophag*):ti,ab,kw	
#3	#1 OR #2	
#4	(wide NEAR/2 area):ti,ab,kw	
#5	(WATS* or 3D?WATS or EndoCDx or Endo?CDx or CDx or brush or transepithelial or trans?epithelial or WATS3D or "WATS-3D" or "WATS 3D"):ti,ab,kw	
#6	(brush NEAR/2 biops*):ti,ab,kw	
#7	MeSH descriptor: [Specimen Handling] explode all trees	
#8	MeSH descriptor: [Biopsy] explode all trees	
#9	MeSH descriptor: [Image Interpretation, Computer-Assisted] explode all trees	
#10	MeSH descriptor: [Electronic Data Processing] explode all trees	

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#11 #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10

#12 #3 AND #11

Ovid MEDLINE(R) and In-Process & Other Non-Indexed Citations and Daily

1 exp Barrett Esophagus/

2 (barrett* esophag* or barrett* oesophag*).ti,ab,kw.

3 1 or 2

4 (wide adj2 area).ti,ab,kw.

5 (WATS* or 3D?WATS or EndoCDx or Endo?CDx or CDx or brush or transepithelial or trans?epithelial or WATS3D or "WATS-3D" or "WATS 3D").ti,ab,kw.

6 (brush adj2 biops*).ti,ab,kw.

7 exp specimen handling/ or exp biopsy/

8 exp image interpretation, computer-assisted/ or Automatic Data Processing/

9 7 or 8

10 4 or 5 or 6 or 9

11 exp Esophagus/

12 (esophag* OR oesophag*).ti,ab,kw.

13 11 or 12

14 3 and 10 and 13

15 sensitiv:.mp.

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16 diagnos:.mp.

17 di.fs.

18 or/15-17

19 animals/

20 humans/

21 19 not (19 and 20)

22 18 not 21

23 14 and 22

Filter: Haynes RB, Wilczynski NL. Optimal search strategies for retrieving scientifically strong studies of diagnosis from Medline: analytical survey. BMJ. 2004 ;328(7447):1040. Also at http://hiru.mcmaster.ca/hiru/HIRU_Hedges_MEDLINE_Strategies.aspx

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PubMed

***

"Barrett Esophagus"[Mesh] OR barrett esophag*[tiab] OR barrett's esophag*[tiab] OR barretts esophag*[tiab] OR barrett oesophag*[tiab] OR barrett's oesophag*[tiab] OR barretts oesophag*[tiab]

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wide-area[tiab] OR WATS*[tiab] OR WATS[tiab] OR "3D WATS"[tiab] OR "3D-WATS"[tiab] OR WATS3D[tiab] OR WATS-3D[tiab] OR "WATS 3D"[tiab] OR EndoCDx[tiab] OR Endo-CDx[tiab] OR CDx[tiab] OR brush[tiab] OR transepithelial[tiab] OR trans-epithelial[tiab] OR "trans epithelial"[tiab] OR "wide area"[tiab] OR ("Specimen Handling"[Mesh:NoExp] OR "Biopsy"[Mesh] OR "Automatic Data Processing"[Mesh:NoExp] OR "Image Interpretation, Computer-Assisted"[Mesh])

("Esophagus"[Mesh] OR esophag*[tiab] OR oesophag*[tiab])

sensitiv*[tiab] OR sensitivity and specificity[Mesh] OR diagnos*[tiab] OR diagnosis[MeSH:noexp] OR diagnostic* [MeSH:noexp] OR diagnosis,differential[MeSH:noexp] OR diagnosis[Subheading:noexp]

#1 AND #2 AND #3 AND #4

NOT ("Animals"[Mesh] NOT ("Animals"[Mesh] AND "Humans"[Mesh]))

Embase (Elsevier)

***

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(barrett* NEXT/1 esophag*):ab,ti OR (barrett* NEXT/1 oesophag*):ab,ti OR 'Barrett esophagus'/exp

WATS*:ab,ti OR WATS:ab,ti OR '3D WATS':ab,ti OR '3D-WATS':ab,ti OR WATS3D:ti,ab OR WATS-3D:ti,ab OR "WATS 3D":ti,ab OR EndoCDx:ab,ti OR Endo?CDx:ab,ti OR CDx:ab,ti OR (brush NEAR/2 biops*):ab,ti OR transepithelial:ab,ti OR trans?epithelial:ab,ti OR ('sampling'/exp OR 'biopsy'/exp) AND ((computer NEAR/1 assist*):ab,ti OR (3?dimensional:ab,ti OR three?dimensional:ab,ti OR 3?D:ab,ti OR 3D:ab,ti OR threedimensional:ab,ti) OR 'computer analysis'/exp)

'Esophagus'/exp OR esophag*:ab,ti OR oesophag*:ab,ti

#1 AND #2 AND #3

Web of Science (Clarivate Analytics)

Science Citation Index Expanded (SCI-EXPANDED) --1900-present

Conference Proceedings Citation Index- Science (CPCI-S) --1993-present

Search run June 2019

184 Records

***

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1 TS=("barrett esophag*" OR "barrett's esophag*" OR "barretts esophag*" OR "barrett oesophag*" OR "barrett's oesophag*" OR "barretts oesophag*")

2 TS=(wide NEAR/2 area)

3 TS=(brush NEAR/2 biops*)

4 TS=(computer NEAR/2 assist*)

5 TS=(WATS* OR WATS OR "3D WATS" OR "3D-WATS" OR WATS3D OR WATS-3D OR "WATS 3D" OR EndoCDx OR Endo-CDx OR transepithelial OR trans-epithelial OR "trans epithelial" OR 3-dimensional OR three-dimensional OR 3-D OR 3D)

6 #2 OR #3 OR #4 OR #5

7 TS=(esophag* OR oesophag*)

8 # 1 AND #6 AND #7

ClinicalTrials.gov

***

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(barrett's OR barrett OR barretts) AND (esophagus OR oesophagus OR esophageal OR oesophageal) | Phase 2, 3, 4

World Health Organization International Clinical Trials Registry Platform (WHO ICTRP)

Search run June 2019

47 Records

***

Condition: (barrett's OR barrett OR barretts) AND (esophag* OR oesophag*) | Phase 2, 3, 4

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Supplementary material



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	4-5
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	6
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8-9
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.8-9	8-9
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	10
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	8-10
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7-9
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	10-11
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	10-11
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	9-11
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	10
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	11
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9-11

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Supplementary material



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	15
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	15
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supp table 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	11-13
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	11-15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	11-15
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	15
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	15
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	14-15
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	11-12
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	15
	23b	Discuss any limitations of the evidence included in the review.	18-20
	23c	Discuss any limitations of the review processes used.	18-20
	23d	Discuss implications of the results for practice, policy, and future research.	20-21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	NA
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	7
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	None
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	2
Competing interests	26	Declare any competing interests of review authors.	2
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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